

FINAL PROJECT REPORT



Transmon Gate Optimisation

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Abstract

This project investigates the optimisation of single-qubit X gates for transmon qubits using advanced pulse engineering techniques. By modelling a three-level transmon system and simulating quantum dynamics with the QuTiP framework, we evaluate the fidelity and leakage performance of various pulse shapes, including Gaussian, DRAG, GRAPE-like, FAST DRAG, and HD DRAG. The study compares single-, two-, and three-pulse gate protocols, highlighting the trade-offs between fidelity and spectral leakage. Notably, three-pulse schemes using HD DRAG and FAST DRAG pulses achieved fidelities exceeding 99.6% with minimal population leakage. IQ-plane visualizations further validated the accuracy of state preparation and read-out. These results emphasise the critical role of pulse shaping in high-fidelity gate design and lay the groundwork for future implementation on superconducting quantum hardware platforms.

Introduction

This project focuses on the optimization of the single-qubit X gate, a fundamental operation in quantum computing. The study begins with the theoretical modelling of a qubit-resonator system using the Jaynes–Cummings Hamiltonian, which captures the essential dynamics of a two-level system coupled to a single-mode quantized field.

Basic simulations were performed using the QuTiP library to visualize qubit state evolution and Bloch sphere dynamics. Various standard single-qubit gates were then simulated using QuTiP to develop familiarity with their behaviour.

The primary focus of the project is the X gate. Multiple pulse shapes—Gaussian, Cosine, Lorentzian, DRAG, HD DRAG, and FAST DRAG—were employed to implement the gate, and their performance was analysed in terms of leakage to higher energy levels and gate fidelity. To further investigate implementation strategies, the X gate was also realized using two sequential $\pi/2$ pulses and extended to a three-pulse protocol.

Finally, the optimised X gate was tested on Rigetti’s Quantum Virtual Machine (QVM), using idealised circuit-level simulations. While the QVM emulates the behaviour of superconducting qubit architectures, pulse-level access was not available, and the assessment was therefore limited to gate-level performance under idealised conditions.

Existing Work

This section reviews prior work relevant to the optimization of single-qubit X gates, with a focus on control pulse engineering for transmon qubits. Key contributions include the development of Gaussian and DRAG pulse strategies, as well as more advanced pulse shaping techniques such as FAST DRAG and HD DRAG.

Modelling of Transmon Hamiltonians (Qi Ding, 2023)

Qi Ding [1] provided a comprehensive modelling of transmon systems and pulse design strategies for high-fidelity gates. Though primarily focused on two-qubit interactions, the work details the structure of the transmon Hamiltonian, including contributions from Josephson junction nonlinearity and capacitive coupling. The thesis also highlights the importance of tailoring pulse envelopes (e.g., Chebyshev, Slepian, Kaiser) to control leakage and phase errors. Such modelling underpins the development of optimized single-qubit gates like the X gate.

Simple Pulses for Leakage Elimination (Motzoi et al., 2009)

The DRAG (Derivative Removal by Adiabatic Gate) technique was introduced by Motzoi et al. [3] to mitigate leakage errors in weakly anharmonic qubits. In this approach, a second control channel proportional to the time derivative of the in-phase envelope is introduced, which cancels off-resonant excitation to higher energy levels. DRAG is particularly effective for systems like transmons, where fast Gaussian pulses alone introduce significant leakage. The technique allows for smooth, hardware-friendly implementation and achieves an order of magnitude improvement in gate fidelity for short gate durations.

Gaussian Pulse Engineering for Frequency Selectivity (Bauer et al., 1984)

Gaussian-shaped pulses have long been used in NMR and quantum control due to their frequency selectivity and minimal spectral sidelobes. Bauer et al. [4] demonstrated that Gaussian envelopes produce excitation profiles with negligible sidelobes, reducing the likelihood of off-resonant excitation. In the time domain, the smooth rise and fall of the envelope lead to Bloch trajectories that return the magnetization vector near the initial axis even for large detuning. These features make Gaussian pulses an effective base envelope for DRAG implementations in superconducting qubits.

FAST and HD DRAG (Hyppä et al., 2024)

Hyppä et al. [5] introduced two analytical techniques—FAST DRAG (Fourier Ansatz Spectrum Tuning DRAG) and HD DRAG (Higher Derivative DRAG)—to push the limits of gate speed and fidelity. These methods generalize the DRAG concept by enabling more flexible frequency-domain suppression of leakage transitions. FAST DRAG designs control pulses via cosine series whose spectral energy is minimized over specified frequency intervals. HD DRAG extends the envelope to include higher-order derivatives for greater spectral control. Experimental results on transmon qubits showed leakage errors as low as 3×10^{-5} for $RX(\pi/2)$ gates at 6.25 ns, representing a 20-fold improvement over conventional cosine DRAG pulses.

Theoretical Background

Transmon Qubits

Transmon qubits are a widely used type of superconducting qubit that offer improved coherence by suppressing charge noise. They are based on the Josephson junction but operate in the regime where the ratio of Josephson energy E_J to charging energy E_C is large ($E_J/E_C \gg 1$), which makes the qubit energy levels less sensitive to charge fluctuations.

The Hamiltonian of a transmon is derived from the Josephson junction circuit with a shunting capacitance and can be written as:

$$\hat{H}_{\text{transmon}} = 4E_C(\hat{n} - n_g)^2 - E_J \cos(\hat{\phi}),$$

where $\hat{n}$ is the number operator corresponding to Cooper pair number, $\hat{\phi}$ is the superconducting phase difference across the junction, n_g is the offset charge, $E_C = e^2/2C$ is the charging energy, and E_J is the Josephson energy.

In the transmon regime ($E_J/E_C \sim 50\text{--}100$), the energy levels become nearly harmonic but retain enough anharmonicity to isolate a computational subspace formed by the ground $|0\rangle$ and first excited state $|1\rangle$. This weak anharmonicity allows for fast microwave-driven control while minimizing leakage into higher excited states, especially when shaped pulses (like DRAG or Gaussian) are used.

The key advantage of the transmon design lies in its reduced sensitivity to charge noise and its compatibility with fixed-frequency and flux-tunable architectures. The small energy difference between $|1\rangle$ and $|2\rangle$ levels, however, necessitates careful pulse engineering to prevent leakage, particularly for fast gate implementations.

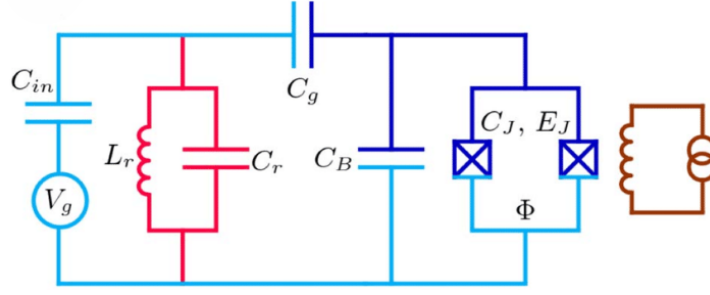


Figure 1: Circuit diagram of a transmon qubit capacitively coupled to a readout resonator. The resonator is formed by an inductor L_r and capacitor C_r , and is driven through the input capacitor C_{in} by a voltage source V_g . The qubit, represented as a Josephson junction with capacitance C_J and Josephson energy E_J , is coupled to the resonator via the gate capacitance C_g . Additional capacitance C_B models shunting or parasitic effects. The external magnetic flux Φ applied through the SQUID loop enables tunability of E_J . [6]

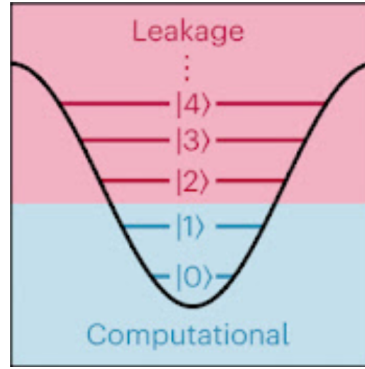


Figure 2: Energy level structure of a transmon qubit. The computational subspace consists of the lowest two energy levels $|0\rangle$ and $|1\rangle$, while higher excited states $|2\rangle, |3\rangle, \dots$ lie in the leakage subspace. [7]

Jaynes–Cummings Model

The Jaynes–Cummings model provides a fundamental description of the interaction between a two-level atom and a single quantized mode of the electromagnetic field. This model forms the basis of understanding cavity quantum electrodynamics (cQED) and is particularly relevant for superconducting qubit systems.

The full Hamiltonian for the driven Jaynes–Cummings model is given by:

$$\hat{H}_{JC} = \hbar\omega_k \hat{a}^\dagger \hat{a} + \frac{1}{2} \hbar\omega_0 \hat{\sigma}_z + \hbar\Omega (\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger) + \hat{H}_{\text{Drive}},$$

where the terms represent:

- $\hat{H}_{\text{field}} = \hbar\omega_k \hat{a}^\dagger \hat{a}$: Energy of the cavity field,
- $\hat{H}_{\text{atom}} = \frac{1}{2} \hbar\omega_0 \hat{\sigma}_z$: Energy of the two-level atom,
- $\hat{H}_{\text{int}} = \hbar\Omega (\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger)$: Atom–field interaction,
- $\hat{H}_{\text{Drive}}$: Represents external driving applied to the system.

Here, $\hat{a}, \hat{a}^\dagger$ are the annihilation and creation operators for the photon field, and $\hat{\sigma}_+, \hat{\sigma}_-, \hat{\sigma}_z$ are the Pauli operators acting on the two-level atom. ω_k is the frequency of the cavity mode, ω_0 is the transition frequency of the atom, and Ω is the atom–field coupling strength.

A schematic representation of the model is shown in Figure 3.

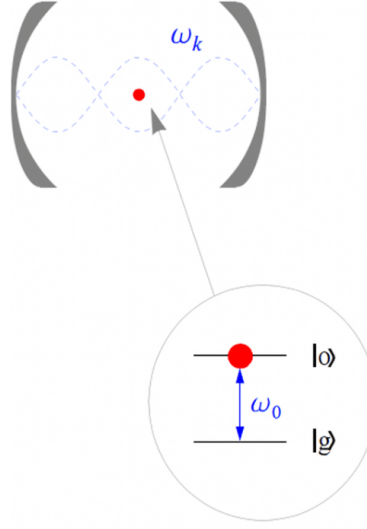


Figure 3: Schematic for the Jaynes–Cummings model. [9]

Types of Drive Pulses

Various pulse shapes and modulation schemes are employed to optimise qubit control, minimize leakage to higher energy levels, and reduce environmental noise susceptibility. Common types include:

- **Gaussian Pulses:** These pulses feature smooth envelopes, which minimize spectral splatter and reduce high-frequency components that could excite unwanted transitions. They are often truncated (e.g., using DRAG pulses) to further improve performance.
- **Derivative Removal via Adiabatic Gate (DRAG) Pulses:** DRAG pulses are specifically designed to suppress leakage to the $|2\rangle$ state in transmon qubits by adding a derivative component to the Gaussian pulse envelope. This compensates for the anharmonicity and improves gate fidelity. The pulse envelope $f_{DRAG}(t)$ is given by:

$$f_{DRAG}(t) = f_{Gaussian}(t) + i\alpha \frac{d}{dt} f_{Gaussian}(t)$$

where α is a parameter related to the anharmonicity of the qubit.

- **Fourier Ansatz Spectrum Tuning (FAST) DRAG Pulses:** This method provides a more flexible control of the frequency spectrum, significantly reducing leakage errors for fast gates without the need for closed-loop optimization. FAST DRAG achieves stronger and wider spectral suppression of the unwanted ef-transition compared to conventional approaches, helping to mitigate leakage to the f-state. It parametrizes the control-pulse envelope using a Fourier cosine series. FAST DRAG can also reduce the bandwidth of the

control pulse, often indirectly lowering the peak amplitude, and its calibration is relatively straightforward as parameters are mostly decoupled.

- **Higher-Derivative (HD) DRAG Pulses:** An alternative pulse-shaping method extending DRAG by incorporating higher-order derivatives to achieve strong spectral suppression of leakage transitions, particularly the ef -transition. This method minimizes the spectrum of the in-phase envelope around the qubit anharmonicity while using standard DRAG to obtain the quadrature component and further suppress leakage. A special case of HD DRAG contains derivatives up to the second order in the in-phase component and up to the third order in the quadrature component. HD DRAG shares common advantages with FAST DRAG, including strong spectral suppression of leakage transitions. HD DRAG can be viewed as a special case of FAST DRAG with specific parameters.
- **Gradient Ascent Pulse Engineering (GRAPE) Pulses:** GRAPE is a numerical optimal control algorithm used to design control pulses for quantum systems to achieve desired operations with high fidelity. It is an iterative algorithm based on the gradient ascent method, maximizing the fidelity between the desired and actual quantum operations by adjusting control parameters (e.g., amplitude and phase of the control fields) over discrete time steps. GRAPE is highly flexible and can be applied to a wide range of quantum systems and objectives, making it a powerful tool for optimizing quantum gates, state preparation, and robustness against noise. The update rule for the control pulse $u(t)$ at each iteration can be generally expressed as:

$$\Delta u(t) = \epsilon \frac{\partial F}{\partial u(t)}$$

where ϵ is the learning rate and F is the objective function (e.g., fidelity) to be maximized.

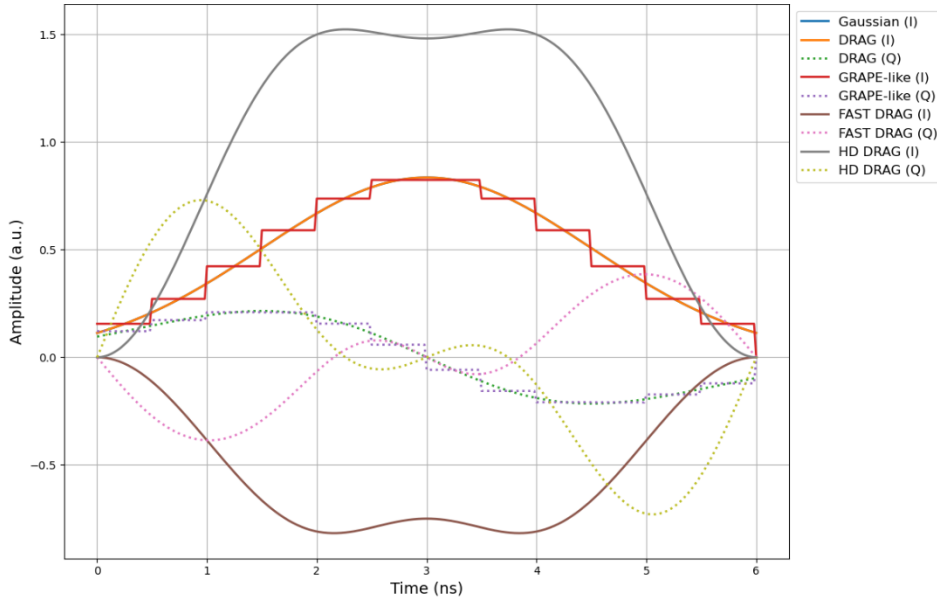


Figure 4: Comparison of I and Q components for various pulse shapes. The plot highlights differences in amplitude profiles and shaping strategies for Gaussian, DRAG, GRAPE-like, FAST DRAG, and HD DRAG pulses

Simulation Methodology

All simulations were performed using the **QuTiP** (Quantum Toolbox in Python) library, along with standard scientific computing libraries such as **NumPy** and **SciPy**. QuTiP is an open-source Python framework specifically designed for simulating the dynamics of open quantum systems. It provides efficient numerical solvers for various quantum master equations, including the Schrödinger and Lindblad equations, and offers tools for manipulating quantum states and operators. Its capabilities were instrumental in accurately modelling the transmon qubit and its interaction with microwave pulses.

Jaynes-Cummings Model and Qubit Dynamics Visualizations

The project began with initial simulations focused on the Jaynes-Cummings model. This involved plotting vacuum Rabi oscillations for a closed quantum system and extending the analysis to an open quantum system by incorporating losses to the environment using collapse operators. The case of a non-zero temperature bath ($T > 0$) was also explored, which activates photon creation by the environment. These implementations were based on examples and guidance from the QuTiP website.

Following these simulations, efforts were directed towards creating animations of qubit dynamics on the Bloch sphere. This started with simple visualizations like Larmor precession and progressed to successfully simulating Rabi oscillations, where the state vector oscillated between the ground and excited states. Animated Bloch-sphere visualizations were also generated to show the evolution of X, Y, Z, and Hadamard gates.

Transmon Qubit Model and Hamiltonian

The transmon qubit was modelled as a three-level quantum system to accurately capture potential leakage to the second excited state, crucial for understanding the performance of shaped control pulses. The computational basis states, $|0\rangle$, $|1\rangle$, and $|2\rangle$, were represented using truncated Fock states. The system Hamiltonian, including time-dependent drive terms in the rotating frame, was implemented using QuTiP’s Schrödinger equation solver, `mesolve`. This allowed for the simulation of coherent qubit evolution under the influence of applied microwave fields.

Control Pulse Implementation and Population Dynamics

Control pulses, including Gaussian, DRAG, FAST DRAG, HD DRAG, and GRAPE pulse shapes, were defined analytically as time-dependent functions and passed to the `mesolve` solver via callable objects.

During each simulation run, the time-evolving state vector $|\psi(t)\rangle$ was computed. The population in each energy level ($P_j(t)$ for $j = 0, 1, 2$) was tracked using the projection operators:

$$P_j(t) = |\langle j|\psi(t)\rangle|^2, \quad j = 0, 1, 2, \quad (1)$$

This allowed for real-time monitoring of excitation and leakage dynamics, providing critical insight into the effectiveness of different pulse shapes in maintaining population within the computational subspace.

Visualization and Sequential Gate Protocols

In addition to single-pulse gate implementations, sequential gate protocols were also studied. This involved applying a sequence of two or three shaped pulses—each carefully designed to enact partial transitions such as $\pi/2$ or $\pi/3$ rotations—to implement a full X gate. The state evolution and leakage were meticulously analysed at each step for both Gaussian and DRAG-based protocols, providing valuable insight into the performance and accumulation of errors in composite pulse sequences.

Simulation Results

Vacuum Rabi Oscillations [8]

The figure shows Vacuum Rabi oscillations, depicting the coherent energy exchange between a two-level atom and a single mode of an optical cavity. The occupation probabilities of the cavity mode and the atom’s excited state oscillate sinusoidally, out of phase, indicating reversible excitation transfer.

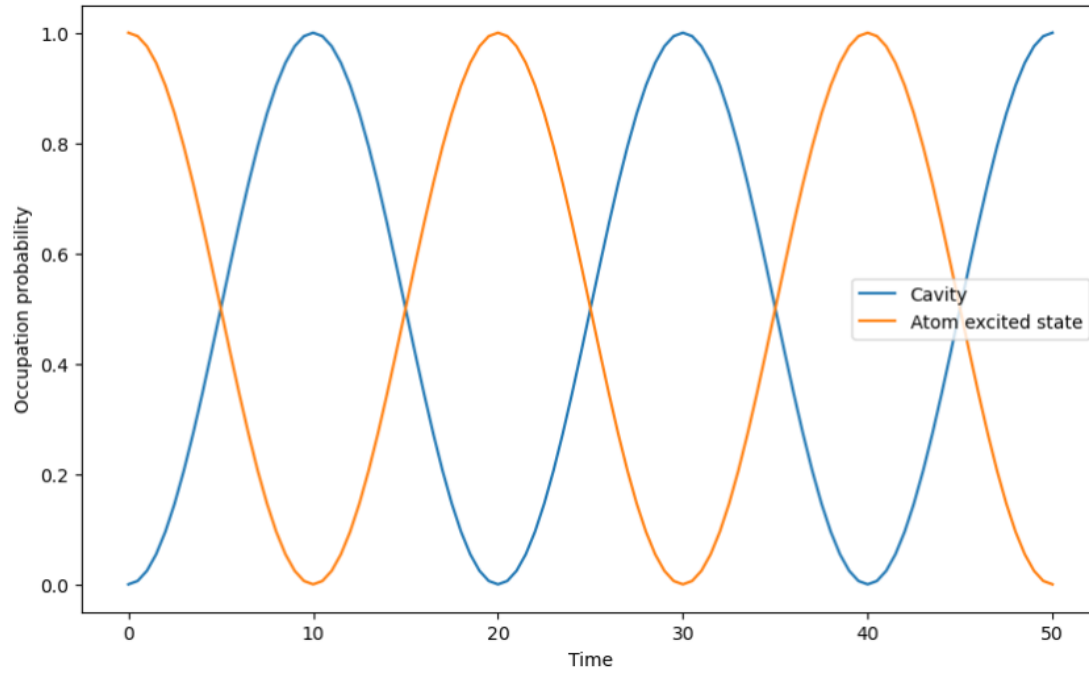


Figure 5: Vacuum Rabi oscillations showing the occupation probabilities of the cavity mode and the atom's excited state as a function of time.

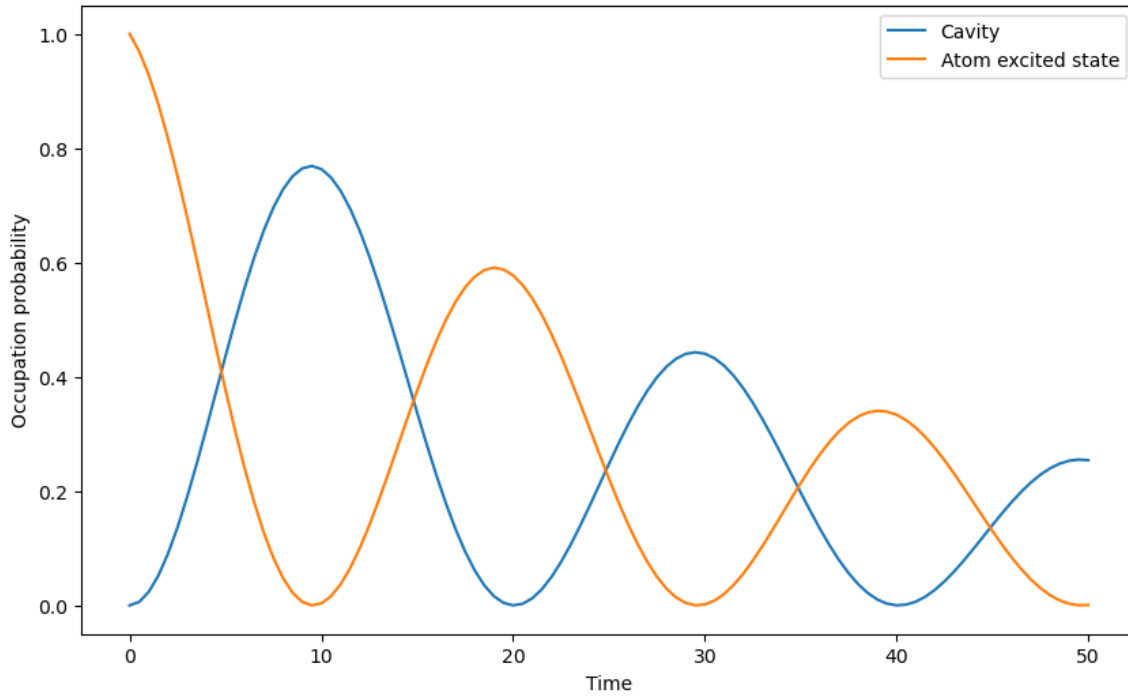


Figure 6: Simulated Rabi oscillations showing the decay of population coherence due to environmental dissipation.

Simple Gate Simulations

Single-qubit quantum gates—X, Y, Z, and Hadamard—were implemented, and the fidelity between the evolving state and the target state was computed at each time interval. The gate operation time was identified as the point at which the fidelity reached its maximum. Animated

Bloch-sphere visualizations were generated to illustrate the state evolution.

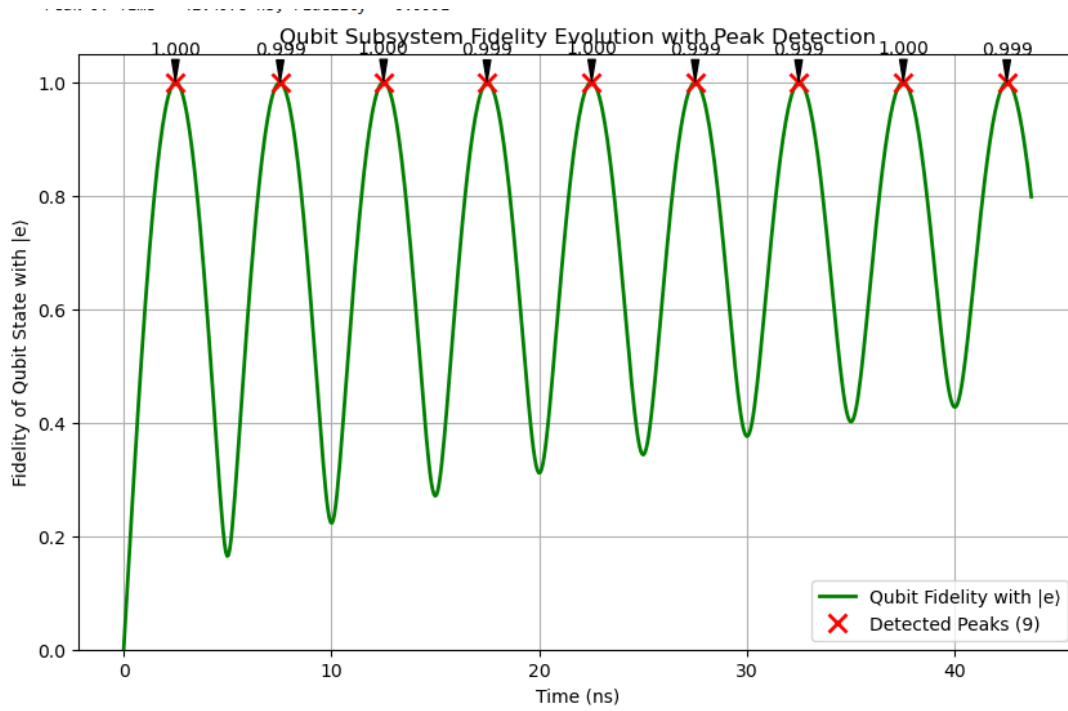


Figure 7: Fidelity - Peak detection for X gate

Bloch Spheres

The Bloch sphere is a geometric representation of a qubit state. The green arrow represents the instantaneous qubit state vector, while the blue line shows the trajectory of the qubit on the Bloch sphere as it evolves under the action of the quantum gate.

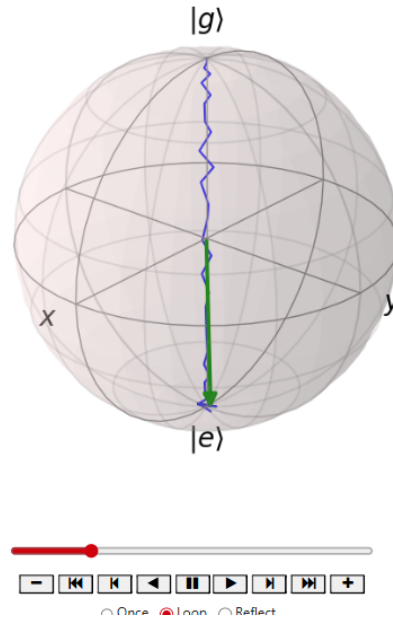
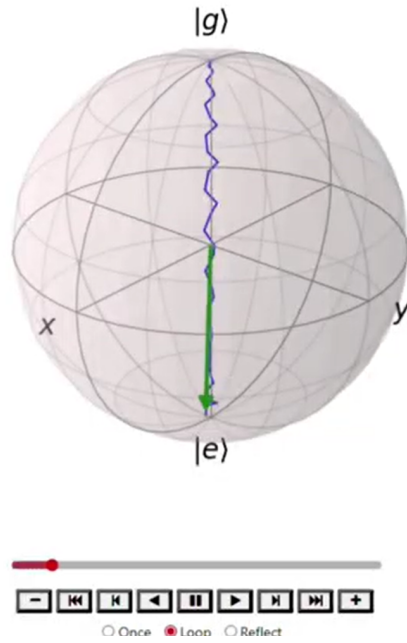


Figure 8: X Gate: The trajectory starts at the ground state $|g\rangle$ (south pole) and moves to the excited state $|e\rangle$ (north pole), corresponding to a rotation of π about the X-axis.



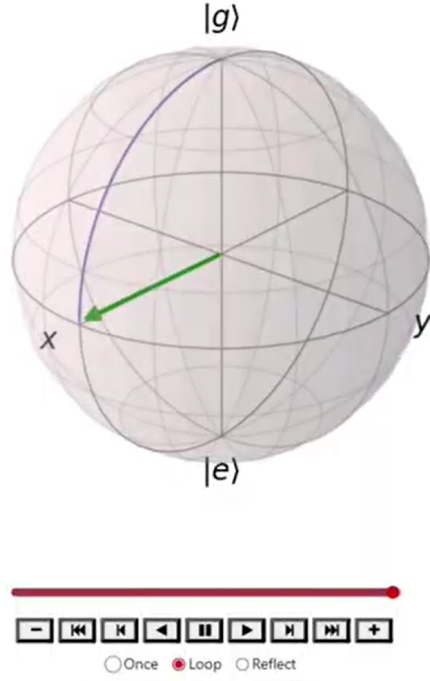


Figure 11: Hadamard Gate: The qubit moves from the ground state to a superposition state on the equator between the X and Z axes, following a diagonal trajectory.

Single X Gate Performance

The standard X gate was first implemented using different pulse shapes on a three-level transmon system. The population transfer from $|0\rangle$ to $|1\rangle$, and leakage to $|2\rangle$, were measured at the end of the gate. Table 1 summarizes the final state populations for each pulse type.

Table 1: One Pulse Implementation

Pulse	Leakage	Fidelity	Gate Time
Gaussian	0.0793 %	97.2346%	6.0 ns
DRAG	0.0766 %	98.3021%	6.0 ns
GRAPE	0.0737 %	98.3324 %	6.0 ns
FAST DRAG	0.0258 %	96.4873 %	6.0 ns
HD DRAG	0.0544 %	95.8323 %	6.0 ns

Each of the following plots displays the time evolution of the population in the three energy levels of the transmon qubit during the X-gate operation. The blue solid line represents the population in the ground state $|0\rangle$, while the orange solid line shows the population in the excited state $|1\rangle$, indicating successful gate action. The green dashed line corresponds to the leakage into the second excited state $|2\rangle$, which is undesirable. The shaded grey area in the background represents the pulse envelope applied during the operation. The ideal behaviour is a smooth transfer of population from $|0\rangle$ to $|1\rangle$ with minimal leakage to $|2\rangle$.

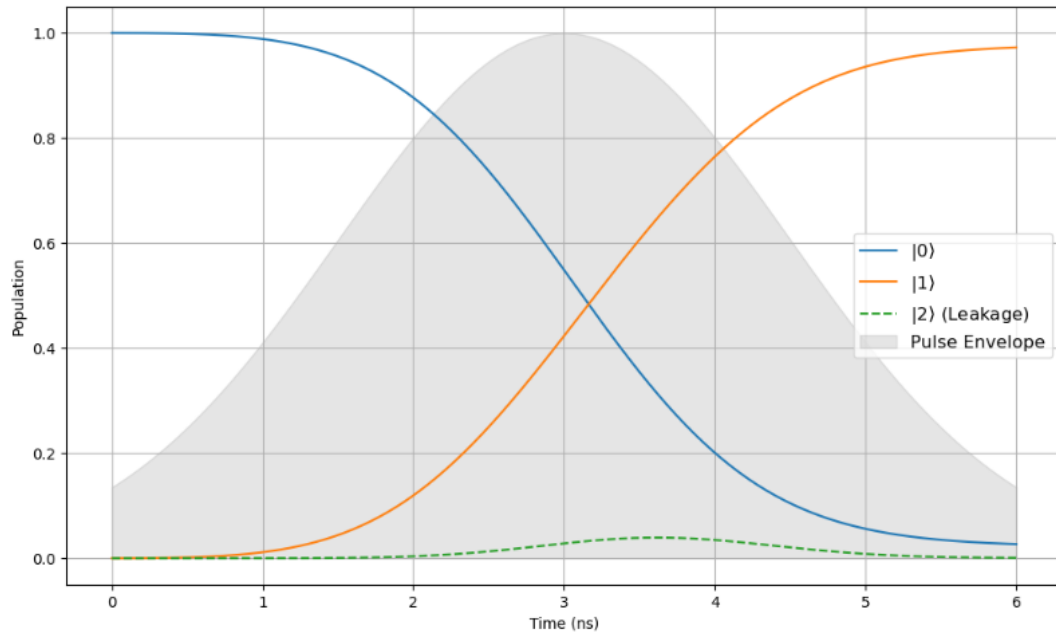


Figure 12: Gaussian Pulse: The population smoothly transfers from state $|0\rangle$ to $|1\rangle$ under a Gaussian-shaped envelope.

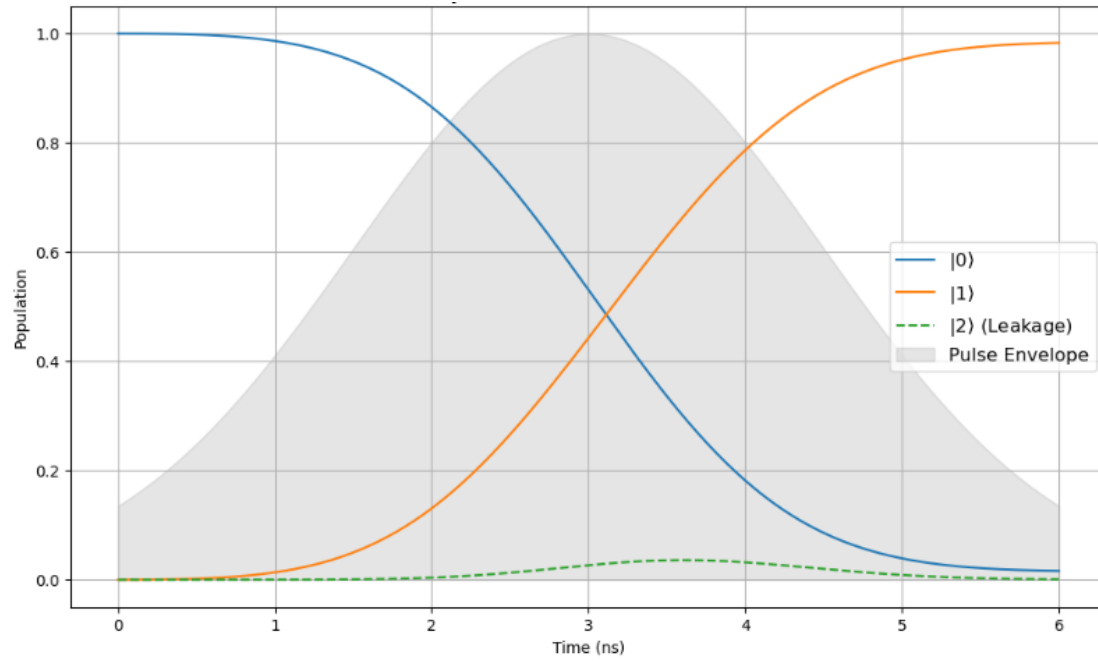


Figure 13: DRAG Pulse: The DRAG (Derivative Removal by Adiabatic Gate) pulse improves transition fidelity and suppresses leakage to $|2\rangle$.

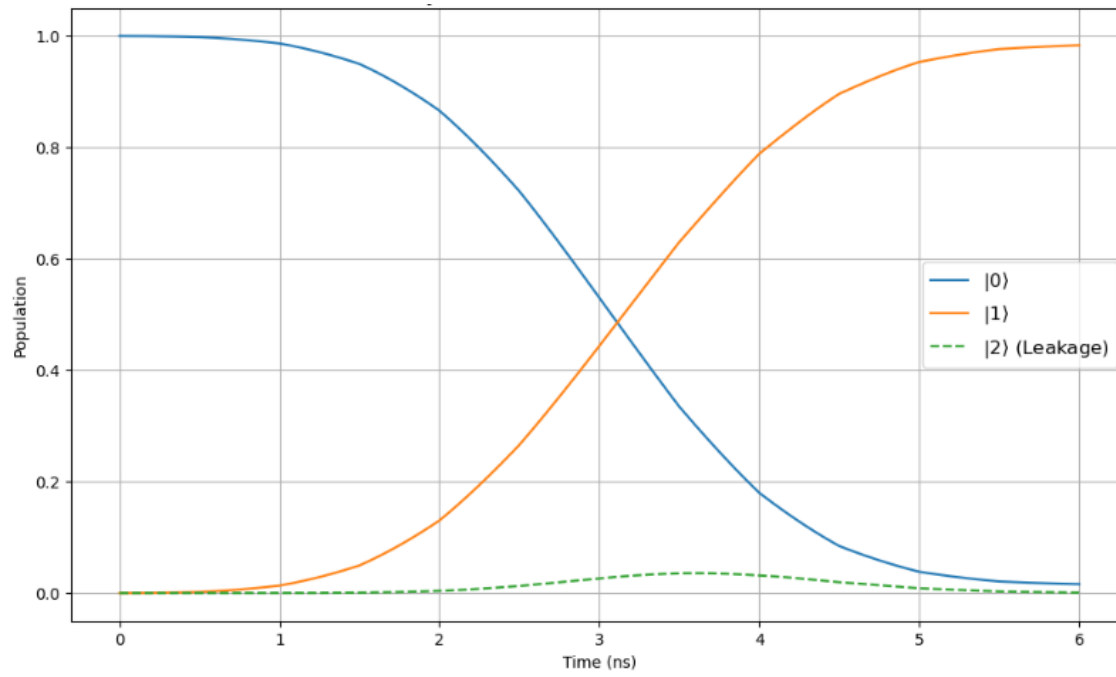


Figure 14: GRAPE-like Pulse: Numerically optimized, maintaining high fidelity with a sharper transition profile.

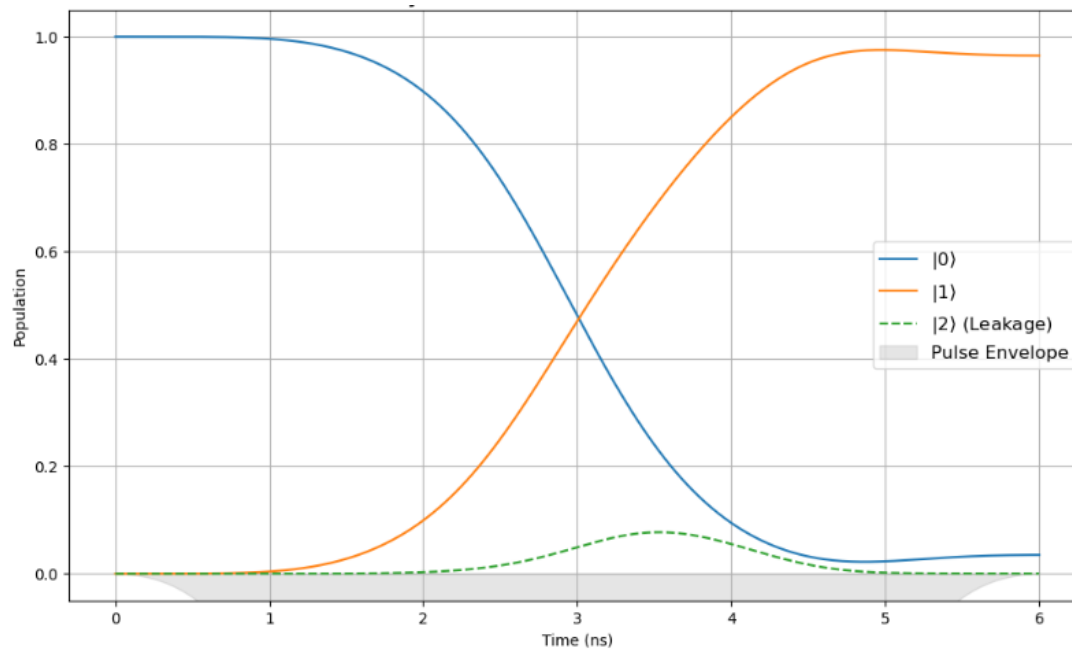


Figure 15: FAST DRAG Pulse: Faster transfer (6 ns) but higher leakage (0.0258%) vs. DRAG; 96.49% in $|1\rangle$.

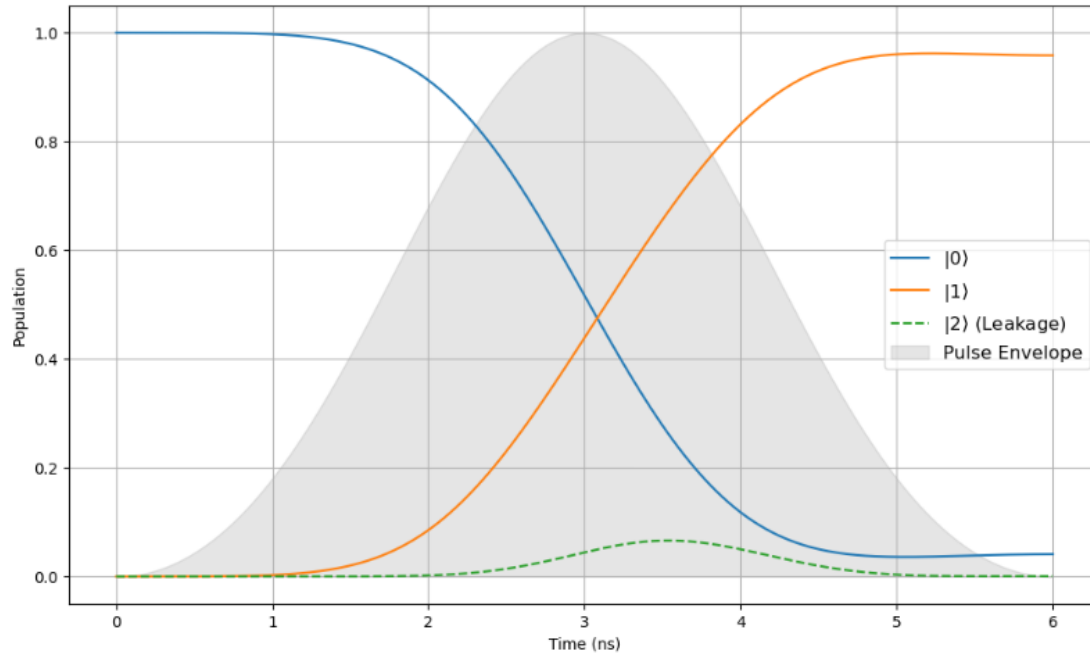


Figure 16: HD DRAG Dynamics: Performs the second best among all pulses, has less fidelity than FAST DRAG

Two-Step X Gate Protocol

To improve control and test pulse robustness, I implemented a two-step gate decomposition:

1. $|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
2. $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \rightarrow |1\rangle$

Table 2: Two-Pulse X-Gate Implementation Results

Pulse	Leakage	Fidelity	Gate Time
Gaussian	0.0120%	98.8978%	12.0 ns
DRAG	0.0098%	99.1728%	12.0 ns
GRAPE-like	0.0087%	99.0158%	12.0 ns
FAST DRAG	0.0003%	89.8133%	12.0 ns
HD DRAG	0.0090%	99.2327%	12.0 ns

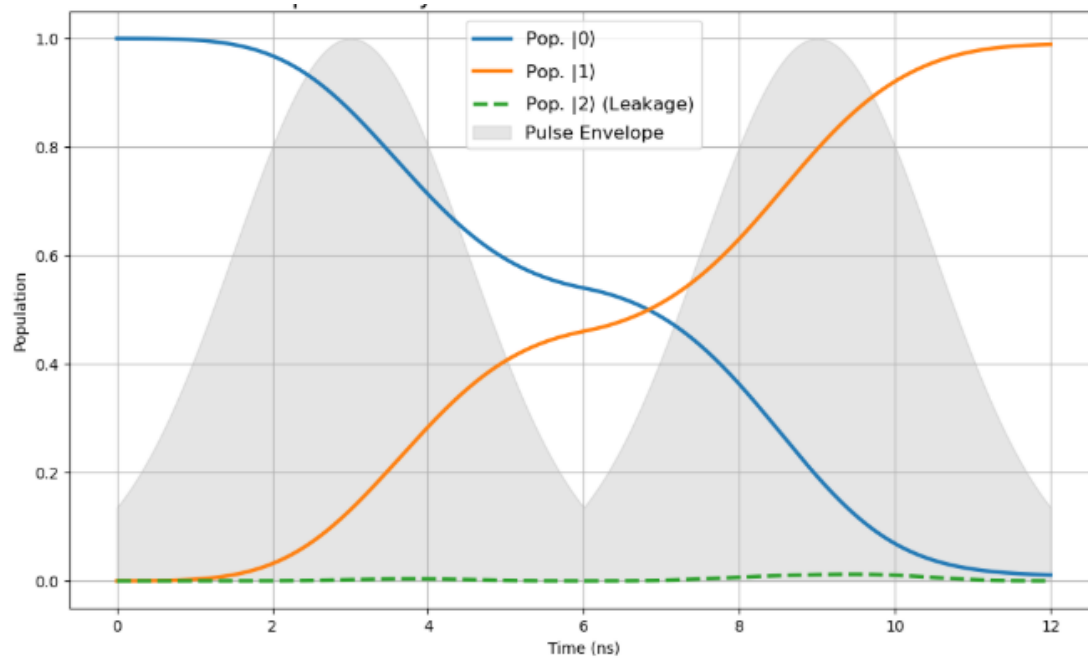


Figure 17: Two-Pulse Gaussian Gate Dynamics: The population smoothly transitions from $|0\rangle$ to $|1\rangle$ with minimal leakage. However, fidelity is slightly reduced.

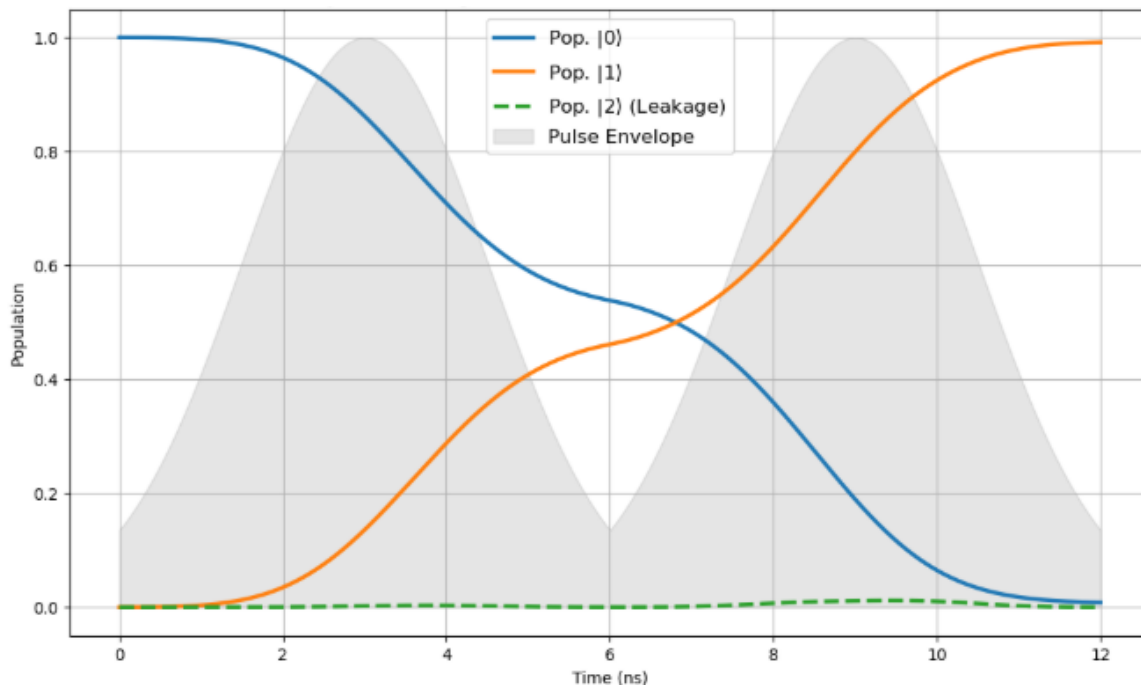


Figure 18: Two-Pulse DRAG Gate Dynamics: Improved transition fidelity compared to Gaussian, with significantly reduced leakage to $|2\rangle$ due to derivative shaping.

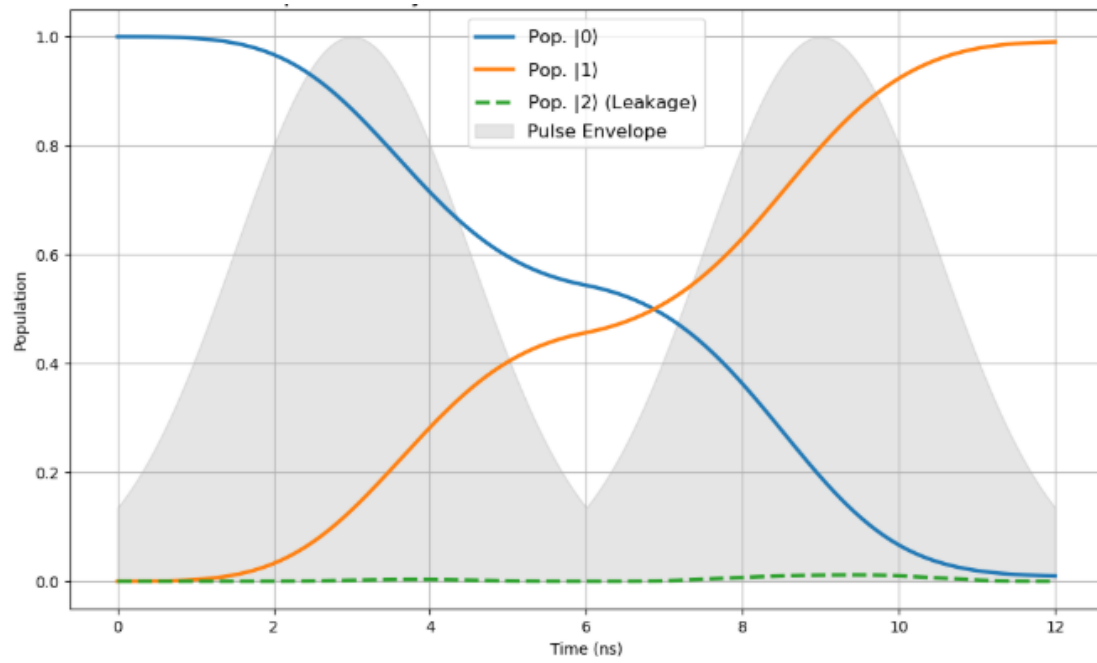


Figure 19: Two-Pulse GRAPE-like Gate Dynamics: This numerically optimized pulse sequence achieves sharper transitions with excellent fidelity and negligible leakage.

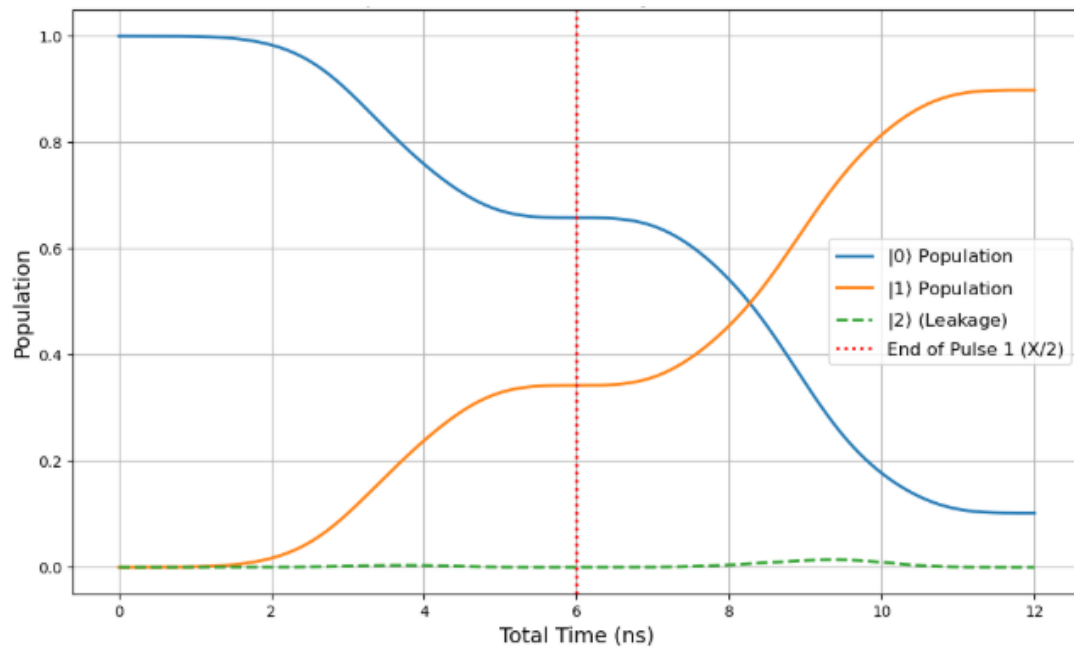


Figure 20: Two-Pulse FAST DRAG Gate Dynamics: Enables faster transfer to $|1\rangle$ in reduced time, but shows a decrease in fidelity.

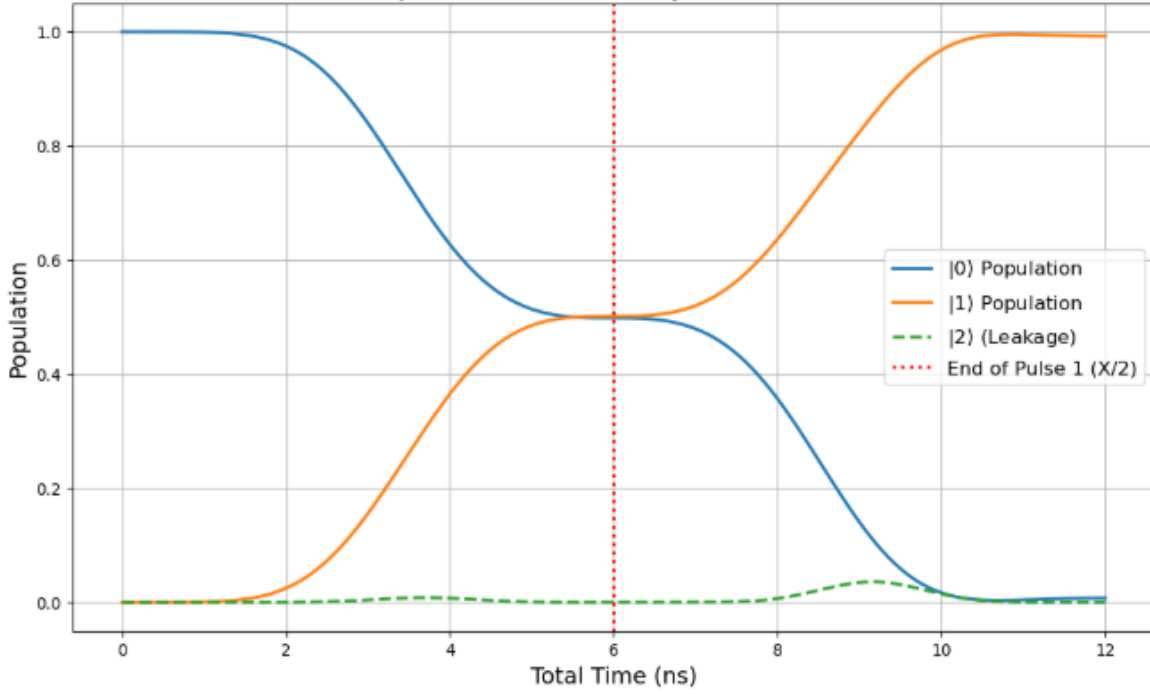


Figure 21: Two-Pulse HD DRAG Gate Dynamics: Among the tested pulse shapes, this achieves the highest fidelity with minimal leakage in two pulse implementation.

To further analyse the gate performance, we also employed IQ (In-phase and Quadrature) readout plots to visualise the simulated measurement results. These plots display clusters of points—each corresponding to a single-shot measurement of the quantum state after applying the two-pulse X gate. The points are colour-coded based on the identified state: blue for $|0\rangle$, orange for $|1\rangle$, and green for $|2\rangle$. Ideally, these measurement outcomes form well-separated blobs in the IQ plane, indicating clear distinguishability between quantum states and minimal readout overlap.

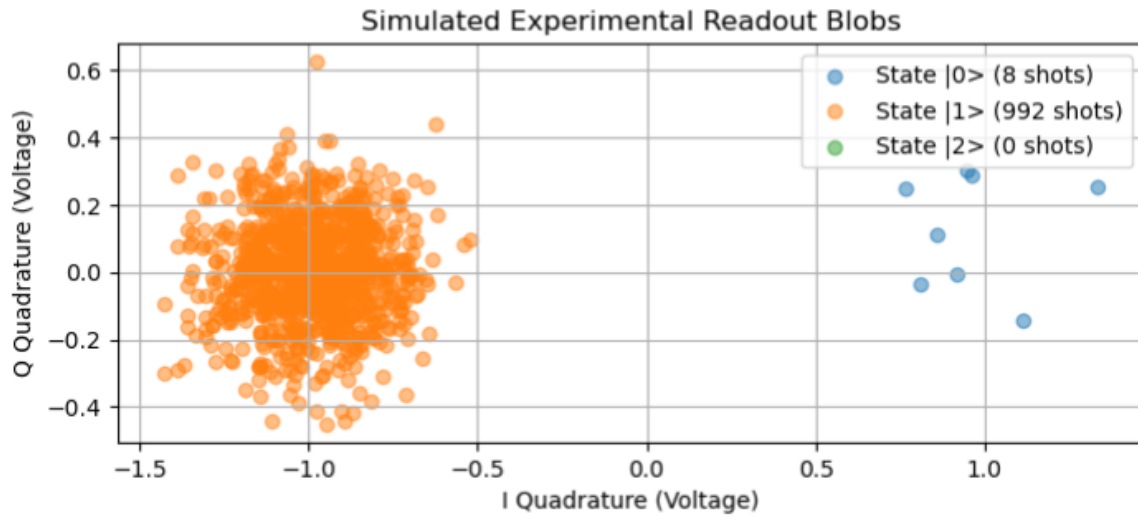


Figure 22: Example of IQ readout plot: Two-Step GRAPE-like Pulse implementation

Three-Step Pulse Implementation

In this approach, the X-gate is implemented using three consecutive pulses, each performing a $\pi/3$ rotation. This fine-grained control allows for smoother transitions and offers potential benefits in minimizing leakage and improving overall gate fidelity.

Table 3: Three-Pulse X-Gate Implementation Results

Pulse Type	Leakage $P(2\rangle)$	Fidelity $P(1\rangle)$	Gate Time
Gaussian	0.0065%	99.1538%	18.0 ns
DRAG	0.0070%	99.2750%	18.0 ns
GRAPE-like	0.0067%	99.1728%	18.0 ns
FAST DRAG	0.0004%	99.6625%	18.0 ns
HD DRAG	0.0039%	99.6040%	18.0 ns

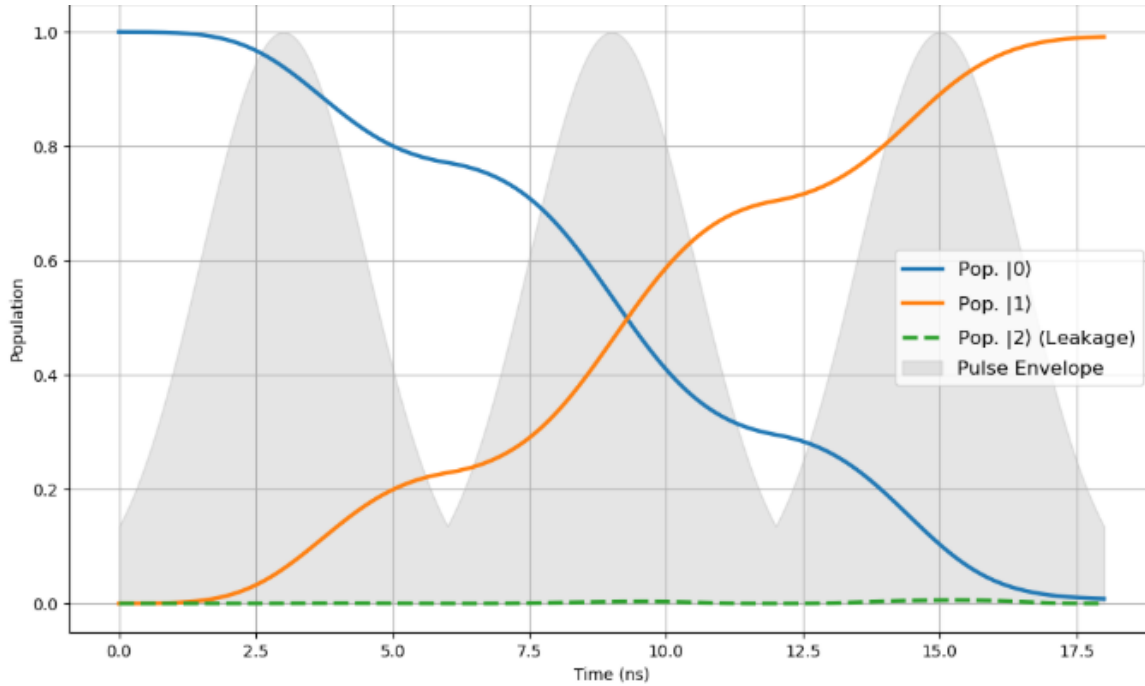


Figure 23: Gaussian Three-Pulse X-Gate Dynamics: Population transfer from $|0\rangle$ to $|1\rangle$ using a Gaussian-shaped pulse sequence, showing a smooth transition with minimal leakage.

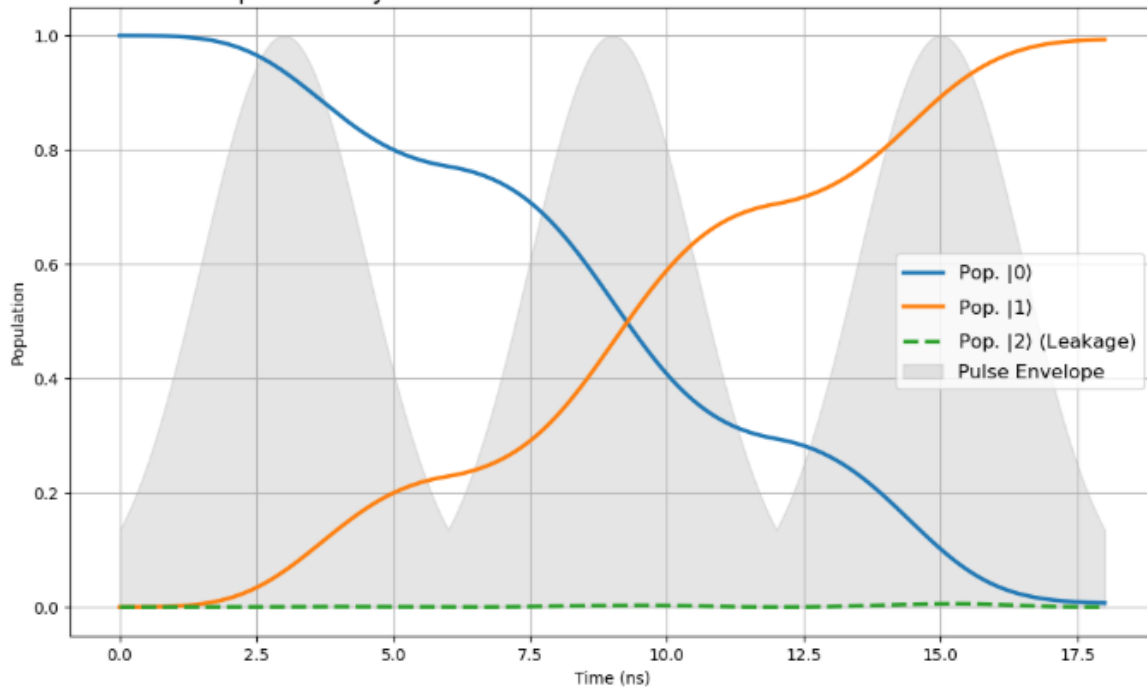


Figure 24: DRAG Three-Pulse X-Gate Dynamics: The DRAG pulse sequence suppresses leakage to $|2\rangle$ while enabling high-fidelity $|0\rangle \rightarrow |1\rangle$ transfer across three pulses.

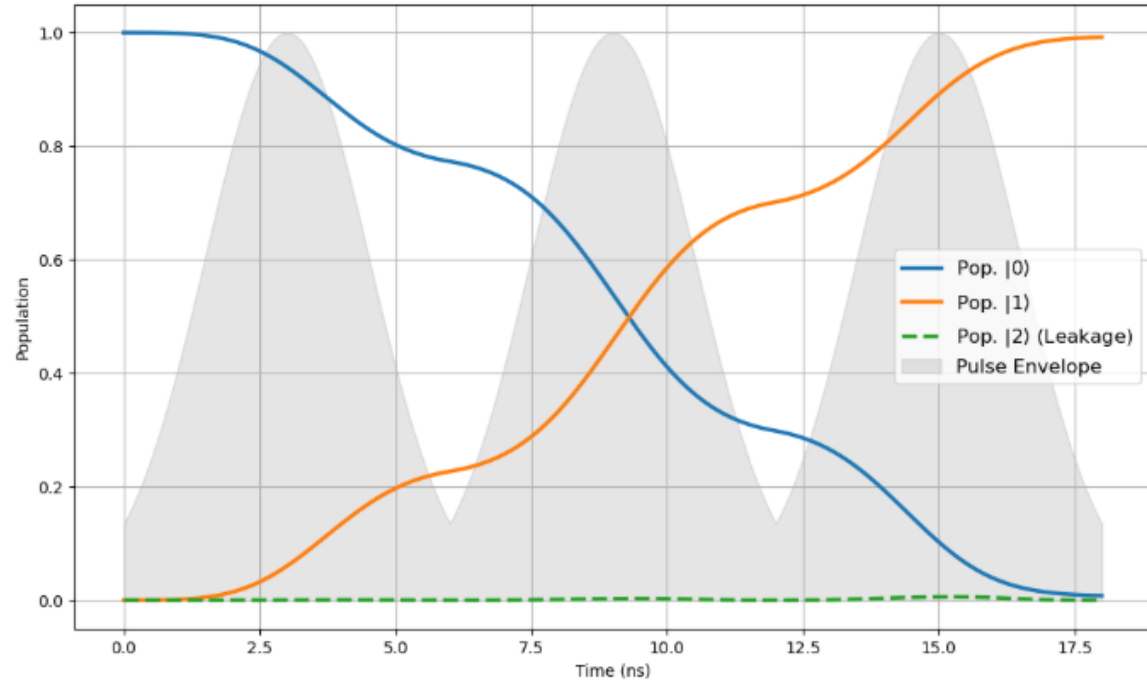


Figure 25: GRAPE-like Three-Pulse X-Gate Dynamics: Numerically optimized GRAPE-like pulses show efficient population inversion with sharp transition regions and negligible leakage.

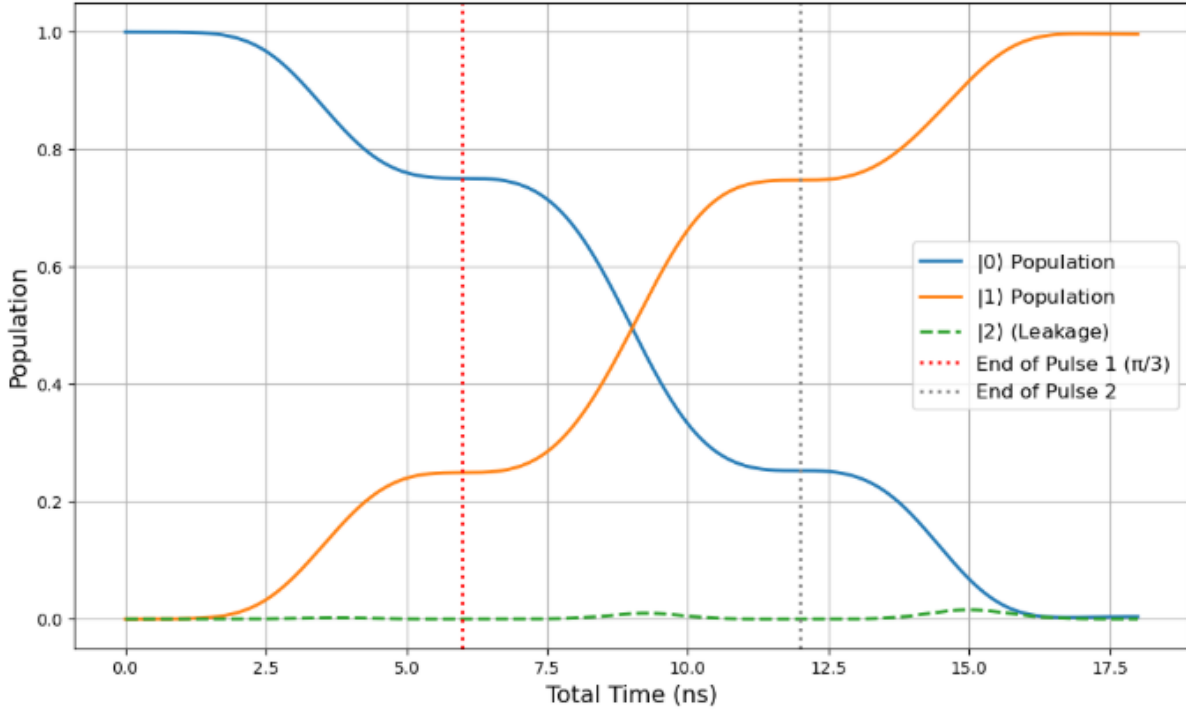


Figure 26: HD DRAG Gate Dynamics: The HD DRAG gate implements a $\pi/3 + \pi/3 + \pi/3$ sequence with minimized leakage and well-separated pulse boundaries for precision control.

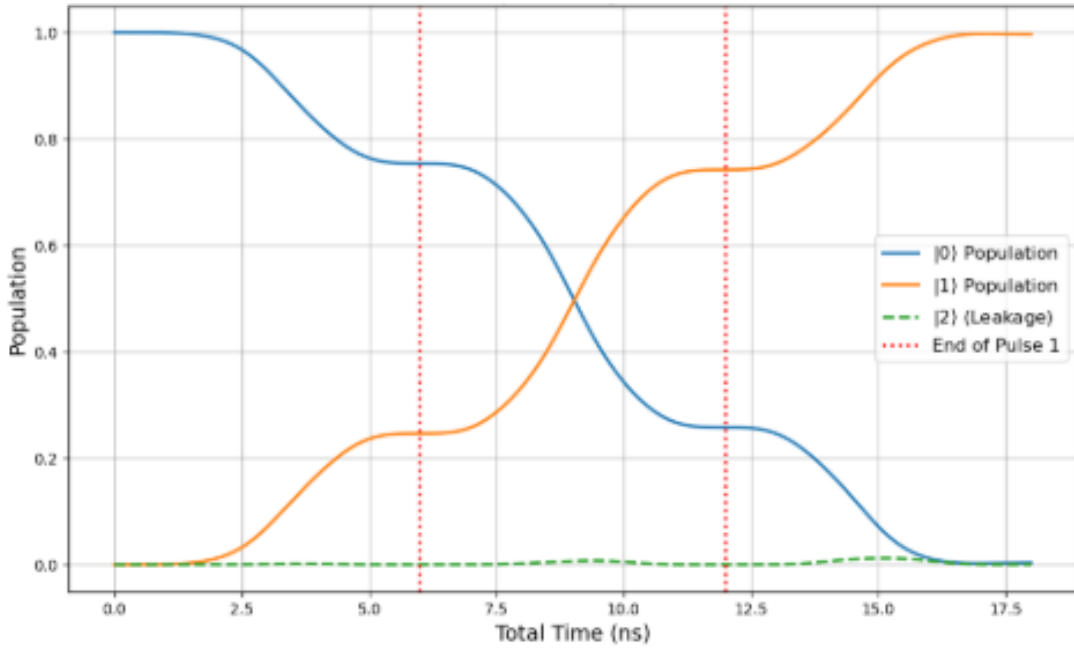


Figure 27: Three-Step FAST DRAG Gate Dynamics (Equal Segments): This symmetric three-step FAST DRAG gate sequence achieves the best performance among all tested methods, enabling near-complete population transfer from $|0\rangle$ to $|1\rangle$ with minimal leakage to $|2\rangle$.

Analysis

The comparative analysis identifies FAST DRAG and HD DRAG as the most effective pulse protocols, each excelling in different aspects of quantum control. FAST DRAG delivers unpar-

alleled leakage suppression, achieving an exceptional leakage rate of just 0.0004% to state $|2\rangle$ - the lowest among all tested protocols. This makes it the optimal choice for applications where minimizing state leakage is absolutely critical.

HD DRAG emerges as the second best, reaching good population transfer to the target state $|1\rangle$ while still maintaining excellent leakage control at 0.0039%.

The three-pulse implementation strategy proves particularly effective for both protocols, demonstrating that additional control degrees of freedom significantly enhance performance compared to simpler pulse sequences.

Simulation Using Rigetti’s Quantum Virtual Machine (QVM)

Rigetti’s Quantum Computing Platform

Rigetti Computing provides a full-stack quantum computing platform featuring both physical quantum processing units (QPUs) and the Quantum Virtual Machine (QVM) simulator. Their superconducting transmon architecture implements gate operations through microwave pulse control, making it an ideal testbed for validating our optimized protocols. While the QVM emulates Rigetti’s hardware behaviour, it provides the flexibility of noise-free testing crucial for protocol verification.

QVM Simulation Constraints and Results

The Rigetti QVM simulations were conducted with several practical limitations that affected the scope of implementation. Pulse-level implementations of both the HD DRAG and FAST DRAG protocols could not be fully realized due to multiple factors, including additional software dependencies that required extensive installation and configuration, as well as time constraints that prevented complete setup. As a result, only basic gate operations were testable within the available resources and time frame.

The simulations that were successfully run demonstrated the expected ideal behaviour of the QVM environment. Both the HD DRAG and FAST DRAG protocols showed identical 100% fidelity in operations, with perfect state transfer evidenced by 5000/5000 measurements in the target $|1\rangle$ state and zero observed leakage to non-computational states. These results clearly indicate that the QVM defaults to perfect simulation behaviour when explicit noise models are not activated, and that the protocol differences observed in the QuTiP simulations were not resolvable at this level of implementation.

Discussion and Conclusion

This project aimed to explore the optimization of the single-qubit X gate for a transmon qubit using a variety of pulse engineering techniques. Through a combination of theoretical modelling, simulation, and experimental insight, the study demonstrated how pulse shaping critically impacts the performance of quantum gates in superconducting circuits.

The simulations began with basic qubit dynamics under the Jaynes–Cummings model and progressed toward more realistic transmon behaviour by considering a three-level system. The

visualization of Bloch sphere dynamics and Rabi oscillations helped build intuition about quantum state evolution under different driving fields. These foundational simulations established confidence in the modelling framework and validated the use of QuTiP for accurate quantum system simulation.

The performance of the X gate was evaluated across three configurations—single-pulse, two-pulse, and three-pulse protocols—using different pulse shapes: Gaussian, DRAG, GRAPE-like, FAST DRAG, and HD DRAG. Among these:

- **Single-pulse implementations** showed that while DRAG and GRAPE-like pulses slightly improved fidelity over Gaussian, FAST DRAG and HD DRAG significantly reduced leakage, although FAST DRAG showed slightly lower fidelity than HD DRAG.
- **Two-pulse protocols** improved fidelity across all pulse shapes by reducing spectral bandwidth per pulse. Notably, DRAG and HD DRAG maintained fidelity above 99% with minimal leakage, and Gaussian pulses also performed surprisingly well in this scheme.
- **Three-pulse implementations** delivered the best overall results. Leakage was reduced to the order of 0.0004% with FAST DRAG, and both FAST DRAG and HD DRAG achieved fidelity greater than 99.6%. These findings support the hypothesis that splitting the gate operation into smaller subrotations can mitigate spectral overlap with noncomputational transitions.

Overall, HD DRAG and FAST DRAG emerged as superior pulse shaping techniques.

Future Work

A natural extension of this work involves implementing the optimized X gate protocols on Rigetti’s Quantum Virtual Machine (QVM) using the PyQuil framework. Since direct access to hardware-level pulse programming through Quil-T requires a paid subscription, initial simulations will focus on incorporating noise modelling into QuTiP by assigning expected error rates for each pulse based on available device characteristics. This approach will allow for realistic performance estimation without immediate hardware access.

Before proceeding to hardware execution, it is essential to understand the specifications and capabilities of Rigetti’s superconducting qubit systems to verify whether pulse-level control and modelling are fully supported. Upon gaining access to the necessary resources, the most effective pulse sequences developed in this study—particularly the FAST DRAG and HD DRAG three-pulse protocols—will be tested on Rigetti’s QVM and, eventually, on real hardware such as the Aspen series.

Future work will also broaden the scope of gate benchmarking beyond fidelity and leakage metrics. This includes analysing gate durations, spectral content, robustness to parameter drifts, and sensitivity to noise, thereby guiding the design of gates that are not only accurate but also resilient and efficient. The ultimate aim is to develop a comprehensive pulse optimization framework for high-fidelity, low-leakage, and fast gate operations tailored to specific quantum hardware constraints.

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